EVORA-AI: A Framework-Adaptive Evidence Assurance Layer for Public-Sector AI Governance

Status: Candidate editorial draft for author review — **not a submission version**.
Source: Proposed reorganization of EVORA-AI v5 following INCISCOS V 2026 reviews.
Scope: Rewritten narrative and core architecture; previous documentary tables and full reference list must be reconciled and audited before assembling the camera-ready paper.
Author decision pending: whether to remove L1–L6 as a normative lifecycle and retain SAIA exclusively as a lightweight assessment and assurance-record process.

Abstract

Public-sector artificial intelligence (AI) governance instruments increasingly define risk, accountability, documentation, and oversight requirements, but their implementation depends on evidence produced across heterogeneous engineering and administrative workflows. Established assurance methods already provide claim–argument–evidence semantics, while recent initiatives address machine-readable evidence and continuous monitoring. The remaining design problem is how to instantiate these capabilities without imposing another governance lifecycle or substantial documentation overhead on resource-constrained public-sector AI teams. This paper refines EVORA-AI as a framework-adaptive assurance layer that binds external requirements to explicit claims, controls, versioned evidence, sufficiency criteria, accountable reviewers, and change-triggered decisions. The Simplified Algorithmic Impact Assessment (SAIA) supplies a lightweight, structured record of those relationships, while evidence is preferentially referenced or harvested from existing engineering tools. A purposive comparison of seven governance environments and ten assurance approaches motivates the proposed integration profile. An illustrative retrieval-augmented generation (RAG) service shows how a claim can be traced to a governance requirement and a proposed acceptance criterion. The contribution is a design-level specification rather than an implemented or field-validated system. The proposed approach is intended to support proportionate, auditable assurance without replacing the institution's governance framework or automating accountable human judgment.

Index terms: AI assurance; public-sector AI governance; evidence traceability; assurance cases; risk proportionality; machine-readable evidence.

I. INTRODUCTION

Public-sector AI can shape access to essential services, allocation of benefits, eligibility decisions, inspection, enforcement, and other administrative outcomes. These uses operate within relationships that differ from ordinary discretionary commercial interactions: individuals may have limited ability to avoid public procedures or choose an alternative provider. As a result, consequential public-sector AI requires more than general statements of responsible use. Governance decisions should be supported by relevant, reconstructable evidence and identifiable decision authority. The type and intensity of assurance must nonetheless follow the applicable legal regime and risk classification rather than a universal set of obligations invented by a new framework [@cadirective; @omb2521; @euaiact].

The practical difficulty is compounded by the resources available for implementation. Public agencies and their development teams must respond to requirements concerning privacy, security, accessibility, transparency, model performance, and accountability while maintaining operational systems. If each governance instrument generates a separate manual documentation process, assurance may consume time better spent improving the system being assessed. This motivates an assurance approach that reuses artifacts already produced by engineering, testing, monitoring, procurement, and incident-management processes.

Existing approaches partially solve this problem from different directions. SACM and GSN provide representations of claims, arguments, and evidence; AMLAS demonstrates structured assurance for machine-learning systems; NIST AI RMF, governmental AI policies, and AI impact assessment instruments define context-sensitive risk management and decision requirements [@sacm; @gsn; @amlas; @nistrmf]. The contribution proposed here is not another assurance language, lifecycle, or risk taxonomy. It is a minimal operational binding profile that makes external obligations, evidence sufficiency, provenance, accountable decisions, and subsequent invalidation explicit, while allowing several governance schemes and engineering environments to use the same underlying record.

Research question: *How can a lightweight, framework-adaptive assurance layer bind heterogeneous public-sector AI governance requirements to verifiable technical evidence and accountable decisions while keeping assurance proportional to risk and minimizing duplicate manual documentation?*

The paper contributes (i) a targeted comparison of seven governance environments and ten assurance approaches; (ii) a minimal assurance binding profile intended for mapping external requirements to evidence and decision conditions; (iii) a design for risk-adaptive evidence reuse and selective revalidation, organized in SAIA; and (iv) a transparent, non-empirical RAG illustration identifying the information and controls required to evaluate a claim. Operational benefit and effectiveness are not asserted; these remain evaluation hypotheses.

II. COMPARATIVE BASIS AND RESEARCH DESIGN

Two intentionally heterogeneous source sets inform the design: public governance environments and methods or infrastructures for assurance. The first set comprises the Brazilian GuIA/MGI ecosystem, Australian Commonwealth and NSW instruments, the United Kingdom, the United States, Canada, the European Union, and Singapore. The second set comprises established assurance representations, impact and risk management standards, operational testing infrastructures, and emerging evidence-as-code or continuous assurance proposals. These were selected purposively for documented relevance to requirements, evidential support, provenance, review, and change management rather than as an exhaustive or systematic review [@guia; @nistrmf; @euaiact; @cadirective; @sacm; @amlas].

The comparison distinguishes **normative specification**, **operational method**, and **technical implementation**. A governance instrument may already specify documentation, risk-based supervision, or revalidation without providing a reusable machine-oriented binding profile. Conversely, a technical evidence platform may capture trustworthy provenance without determining which evidence is legally or contextually sufficient for a public-sector claim. EVORA-AI is directed at this interface. Documentary capability ratings, where retained, must be supported by explicit cell-level references to frozen source versions; they indicate what the reviewed instruments specify, not actual adoption or field effectiveness.

This paper follows a Design Science Research orientation as a nascent artifact proposition [@gregor; @hevner]. Its present scope includes specification and illustration, not software execution, evaluation with public servants, or measurement of administrative burden. Those activities belong to a subsequent research cycle. The separation is intentional: an internally consistent specification is a prerequisite for an empirical test, but it is not evidence that the artifact improves outcomes.

Editorial integration note: The reduced government and assurance comparison tables should be inserted here only after auditing the v5 classifications; do not claim the full comparative matrix has been independently replicated in this draft.

III. EVORA-AI: FRAMEWORK-ADAPTIVE ASSURANCE

A. Architectural boundary

EVORA-AI is a mediation layer between (1) governance sources, (2) system-specific engineering and operational evidence, and (3) accountable decisions. A governance source may be a law, sectoral policy, organizational procedure, GuIA phase-level requirement, NIST AI RMF profile, impact-assessment result, or an internal risk

criterion. None is required to adopt an EVORA-AI lifecycle. The source retains authority over applicability, wording, approval requirements, and legal consequences; EVORA-AI records mappings to assurance claims and supporting evidence.

A **governance adapter** identifies a requirement and its source version, applicability conditions, risk context, relevant decision point, and responsible authority. A **technical adapter** refers to or captures artifacts from CI/CD, evaluation suites, model registries, dataset inventories, security controls, monitoring, and incident handling. The intermediate assurance profile connects these elements through explicit claims and acceptance criteria. The resulting SAIA record is intended to remain reconstructable over time, including when external requirements or system components change.

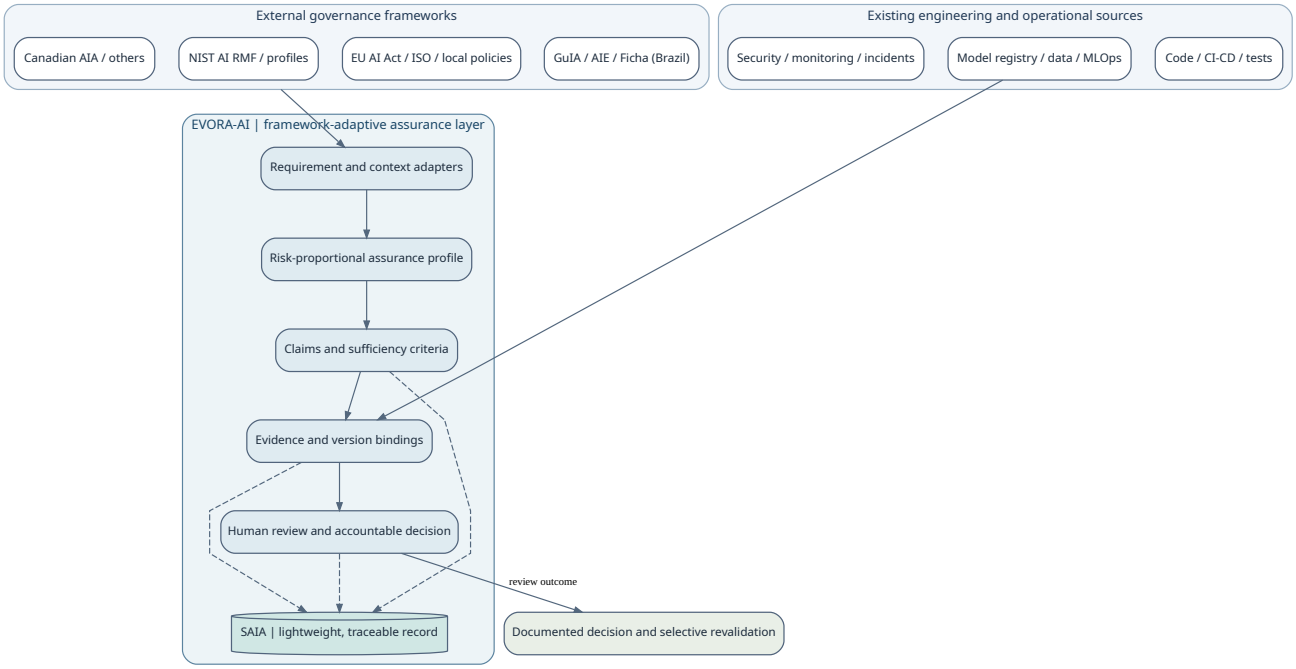


Figure A. Framework-adaptive EVORA-AI architecture, separating external governance, assurance bindings and existing engineering evidence.

B. Minimum assurance contract

For each assessable claim, EVORA-AI specifies the following relationships: external requirement reference; claim identifier and assertion; applicable control; evidence identifier and origin; system/data/model/control version binding; acceptance or sufficiency criterion; evidence status and validity conditions; responsible evidence producer or reviewer; recorded decision and rationale; and triggers for reopening the claim. This is a **minimum semantic contract**, not a replacement for SACM or GSN. The profile must not treat the mere existence of a file as evidence of compliance, nor equate a successful format check with substantive evidence sufficiency.

The proposed invariants are straightforward. An accepted claim must have identifiable supporting evidence and an explicit sufficiency criterion. Evidence must refer to the system configuration for which it was produced. A decision must identify its accountable authority and grounds. A material change affecting an accepted claim must reopen the relevant evidence judgment. A claim with missing, stale, or insufficient evidence must not be silently promoted to accepted status. The exact enforcement mechanisms remain implementation decisions and require subsequent testing.

C. Risk-adaptive assurance profiles

EVORA-AI does not impose a single cross-jurisdictional classification system. Instead, a risk adapter maps the categories and obligations of the governing instrument to a locally approved assurance profile. Lower-impact contexts may require a baseline set of claims, traceable tests, and proportionate review. Higher-impact contexts may require strengthened evidence, qualified human oversight, separation of duties, independent review, and

restrictions on deployment when evidential criteria remain unmet. The source policy or legal regime determines which obligations are mandatory in that context. EVORA-AI can additionally propose more conservative review conditions as explicit artifact-design requirements, but should not present them as universally binding law.

The review model follows consequence, not merely technical complexity. Systems materially affecting rights, benefits, safety, or access to essential public services warrant more demanding justification because affected individuals may have limited choice or effective recourse. This public-law rationale does not justify indiscriminate bureaucracy: increased assurance should attach to the consequential claim and risk, not require every low-risk engineering change to undergo the same review.

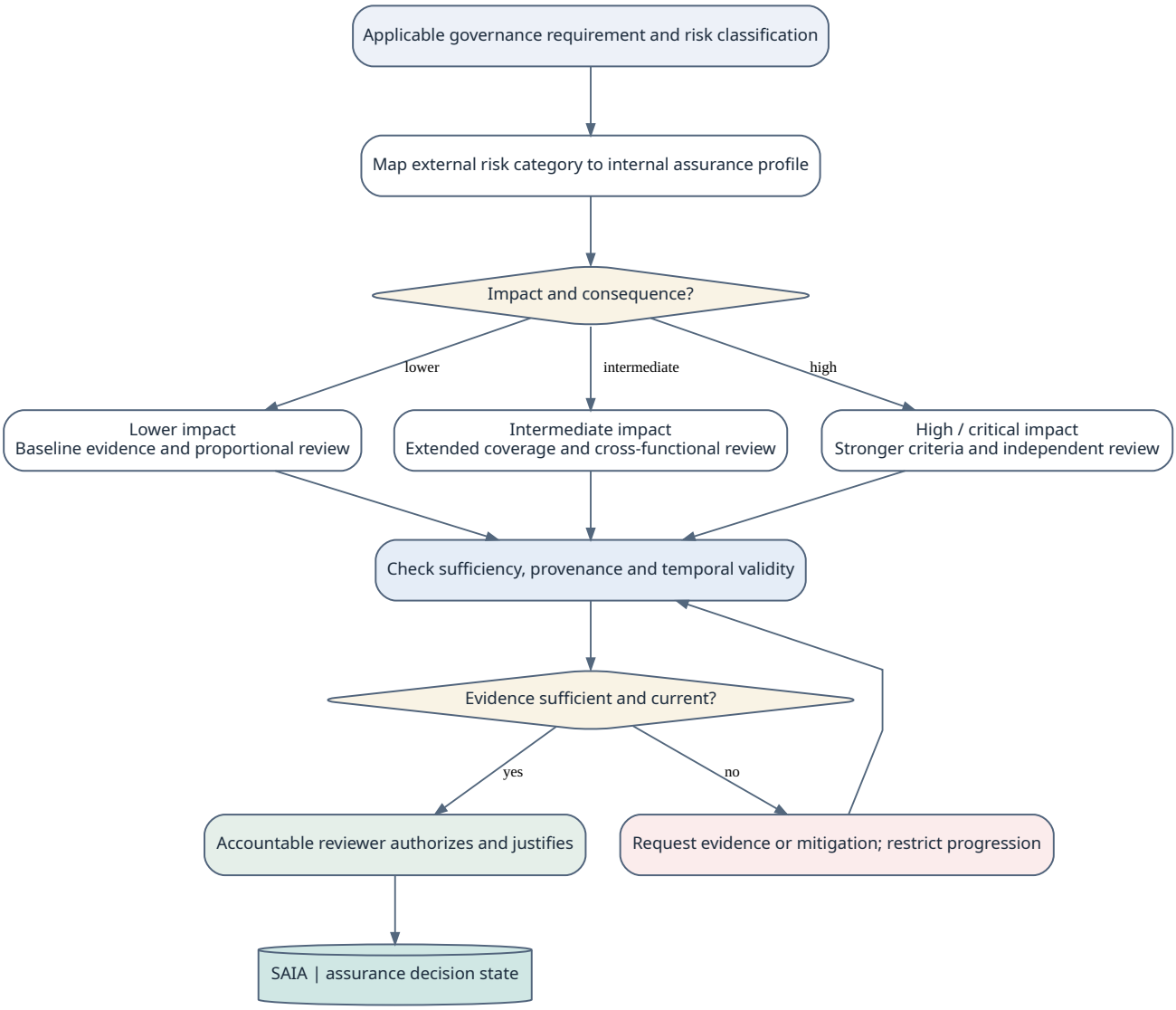


Figure B. Proposed risk-adaptive assurance profiles; external risk categories remain authoritative.

D. Evidence reuse, selective revalidation, and SAIA

Evidence should be referenced from authoritative sources before a team is asked to produce duplicate documentation. Automated retrieval may supply test artifacts, commit or release identifiers, model and dataset versions, monitoring signals, and control configuration states. Rules can determine whether an artifact is present, attributable, linked to the correct version, and within a declared validity period. By contrast, interpretation of contested legal obligations, judgment of substantive sufficiency, risk acceptance, and approval for consequential deployments remain assigned to accountable humans.

Revalidation is triggered by changes in the claim's dependencies: a model update, a material dataset change, a new intended purpose, a new external tool, a revised policy obligation, detected drift, or a security or privacy incident. Rather than invalidating every record indiscriminately, the layer identifies claims potentially dependent on the affected component. Those claims require refreshed evidence or renewed judgment; unaffected records remain available with their original version bindings. SAIA maintains the previous status, the change event, and the new decision. These capabilities are proposed design properties, not claims of implementation.

SAIA retains its approved expansion, **Simplified Algorithmic Impact Assessment**, but its role is tightly bounded. It does not replace an external impact assessment, such as the AIE or a Canadian AIA. It links the relevant risk or assessment outcomes to claims, evidence, and assurance decisions, and maintains a lightweight, structured assurance record. An organization that already uses a system card, AIA, or internal register should be able to map those fields rather than recreate them manually.

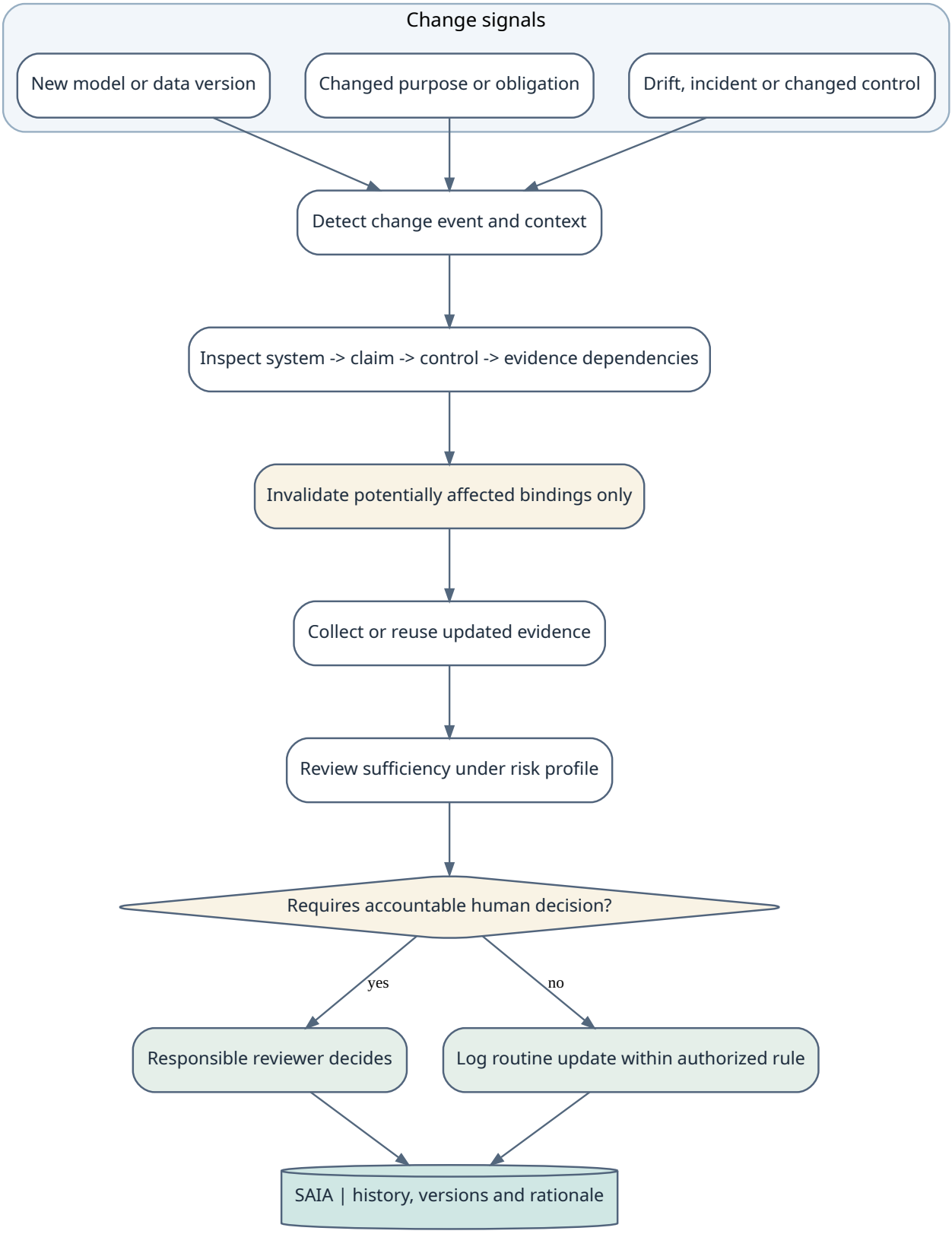


Figure C. Proposed selective revalidation flow, preserving affected evidence bindings and accountable decisions.

IV. ILLUSTRATIVE APPLICATION: PUBLIC-SERVICE RAG

Consider a public agency using a retrieval-augmented generation (RAG) assistant for citizen information services. The illustration addresses a single hypothetical assurance claim so that the required bindings can be inspected rather than inferred from a list of familiar threats. The applicable governance requirement concerns protection of

personal data and authorized access to retrieved institutional content; a GuIA privacy-verification item is one possible Brazilian source [@guia]. The same claim could be attached to another relevant policy or regulatory requirement through a different adapter.

SAIA field	Proposed illustrative record — not executed
External requirement	Applicable privacy/security verification for the public-service RAG system; source and version to be identified
Claim	RAG retrieval and generated answers do not expose documents to unauthorized requester roles within the defined test scope
Control	Role-based retrieval authorization; document classification and access restrictions; adversarial regression tests
Evidence reference	**Pending:** access-control test report, version-bound evaluation logs and access policy snapshot
Sufficiency criterion	Approved test suite covers defined authorization scenarios; no unresolved critical unauthorized disclosure within that tested scope
Subject binding	RAG service release, retrieval index, access-control policy, connected model and dataset versions
Reviewer	Responsible security/privacy reviewer distinct from the test producer where risk profile requires independence
Decision	**Not assessed / pending evidence** — no claim of acceptance or deployment authorization
Revalidation triggers	Retrieval corpus permissions change, model/tool change affecting access behavior, new purpose, or disclosure incident

The example shows what EVORA-AI would require before a reviewer could accept the claim. No measurements, passed tests, verified controls, or performance improvements are asserted. The evidential gap is visible precisely because the proposed record exposes what is not yet available. Additional RAG risks—prompt injection, poisoned documents, hallucination, and vendor dependency—can be mapped using the same contract without generating a new assessment methodology for each threat.

V. DISCUSSION

EVORA-AI's contribution is an integration profile at the boundary between governance requirements and technical evidence. It is not an invention of claim–argument–evidence semantics and does not supersede the policy, lifecycle, or risk taxonomy of the adopting institution. Its design value rests on whether a small set of stable bindings can make otherwise fragmented assurance records comparable, auditable, and revalidatable across systems and frameworks. The GuIA/MGI ecosystem is an important public-sector context for illustrating this proposition, but is not its conceptual prerequisite.

The distinction matters especially in public administration. State-supported AI can affect individuals in ways that are difficult to avoid, reverse, or challenge. This justifies strong evidence and meaningful review for consequential applications, but responsibility cannot be transferred to a software validator: an automatically harvested test log does not determine whether a government may lawfully deploy a system, whether its evidence is substantively sufficient, or whether residual risk should be accepted. Human oversight is part of the assurance contract, not a workaround for incomplete automation.

The second concern is feasibility. Public AI teams frequently operate with restricted technical capacity. For them, an assurance process that requires reconstructing facts already present in code repositories, deployment manifests, test pipelines, and incident systems can weaken engineering productivity. EVORA-AI therefore adopts **evidence reuse before evidence reproduction** as a design objective. The anticipated reduction of manual documentation is an empirical hypothesis to test against a baseline, not a demonstrated outcome.

For evaluation, a subsequent study should compare manual governance documentation and SAIA-assisted workflows on identical cases. Relevant outcomes include the proportion of requirements with evidence meeting explicit criteria, number of stale records detected following change, agreement among qualified reviewers, time needed to prepare and review decisions, hours of manual documentation, and the ability to reconstruct the evidential basis of past approvals. Structural checks of a JSON schema would establish only serialization and basic invariants, not the effectiveness or institutional acceptability of the process.

VI. THREATS TO VALIDITY

The study remains a design-level proposal without field evaluation. The purposive documentary comparison is not exhaustive; ratings and boundaries between governance, assurance methods, and technical infrastructure involve interpretive choices. International instruments differ in binding force, intended audience, and update cycle; their treatment as comparable environments requires explicit source versions and transparent coding rules. The illustrative RAG claim has no executed evidence and cannot demonstrate impact on error rates, administrative effort, review latency, or risk reduction. Finally, the proposed assurance contract may not generalize unchanged to safety-critical systems, autonomous decision systems, or sector-specific legal procedures. These limitations require empirical and cross-framework testing and should not be obscured by architectural claims.

VII. CONCLUSION

The central problem addressed by EVORA-AI is not the lack of AI governance frameworks or the absence of mature claim–evidence methods. It is the difficulty of making those instruments operationally interoperable when public-sector decisions depend on evidence distributed across systems, teams, and moments in time. EVORA-AI proposes a lightweight, framework-adaptive assurance contract that binds an external obligation to an assessable claim, a control, version-specific evidence, a declared sufficiency criterion, and an accountable decision. SAIA provides the evolving record through which those bindings could be maintained, inspected, and reopened after material changes.

The proposal is deliberately narrower than a new lifecycle framework. Governance authorities retain responsibility for requirements, risk classes, and approval rules; engineering systems remain the primary sources of technical evidence. The assurance layer operates between them. This design addresses a specific tension in public administration: the need for stronger justification of high-consequence AI uses, given the State's capacity to affect citizens' rights and access to services, without diverting scarce technical resources into duplicated documentation. Risk-proportional review and preferential reuse of existing evidence are therefore part of the artifact's specification rather than claims of demonstrated efficiency.

The present paper establishes a conceptual architecture and an inspectable demonstration record, not a completed implementation or empirical validation. Future work must test whether mappings across governance regimes preserve the intended claim meaning, whether selective revalidation identifies stale evidence accurately, and whether human reviewers find the resulting records sufficient and usable. Equally important, evaluation must measure whether the process adds or reduces administrative work relative to existing practice. The contribution will be demonstrated only if it supports accountable decisions **and** remains feasible for the teams delivering public AI services.

The intended direction is not governance by accumulating documents. It is governance in which consequential decisions can be traced to the evidence, assumptions, versions, and responsible judgments on which they depended, while routine evidence collection remains as close as possible to the engineering processes that produce it.

Editorial integration and scientific integrity notes

- This is a **candidate restructuring**, not a claimed final or accepted manuscript; changes to L1–L6 and SAIA's functional boundary need the author's ratification.
- The original v5 Tables I and II and bibliography are **not** silently reproduced because their rating/source mappings still require verification against the exact sources and cut-off date.
- The SAIA sample is explicitly **not executed**: it has no fabricated test result, trace, claim acceptance, or deployment approval.
- For a camera-ready version, reconcile v5 BibTeX/`thebibliography` keys with the cited references before LaTeX conversion; verify IEEE A4 page size and conference template.
- Related Mermaid diagrams accompany the report: architecture, risk-profile decision, and selective revalidation.